

Computation of Magic Constants of Distance Magic Graphs

1 Introduction and Results

Throughout this article, we assume that graph $G = (V, E)$ is a finite simple graph with V denoting the set of vertices and E the set of edges. For a vertex, u , its neighborhood is given by $N(u) = \{v \in V : uv \in E\}$. A positive integer k is called a *magic constant* if there is a graph G along with a bijective function $f : V(G) \rightarrow \{1, 2, \dots, |V(G)|\}$ such that the weight of the vertex $w(v) = \sum_{uv \in E} f(u)$, remains constant and equal to k for all $v \in V$. Such a labeling f is called *distance magic labeling*. A graph equipped with distance magic labeling is known as a *distance magic graph*.

The following result is known

Theorem 1.1. *All positive integers except 1, 2, 4, 6, 8, 12, and 16 are magic constants.*

Proof. It was shown in [3] that the integers 1, 2, 4, 6, 8 and 12 are not magic constants. The case for $k = 16$ follows from an algorithm in the paper by Pawar et al. ■

Regardless of how the vertices are labeled, the magic constant, if it exists, remains invariant (see [1, 6]), that is, it is independent of the distance magic labeling.

Let G be a graph. A subset S of the vertex set of G is said to be a *dominating set* if there is a function $f : V \rightarrow \{0, 1\}$, called *dominating function*, such that $f(S) = \{1\}$, $f(V \setminus S) = \{0\}$ and for each vertex $v \in V \setminus S$, $\sum_{u \in N(v)} f(u) \geq 1$. The *domination number* $\gamma(G)$ of a graph is given by

$$\gamma(G) = \min\{|f| : f \text{ is a dominating function}\},$$

where $|f| = \sum_{v \in V} f(v)$. There is another variation of domination called fractional domination.

Definition 1.1. A function $g : V(G) \rightarrow [0, 1]$ is called a *total dominating function* of the graph G if $\sum_{u \in N(v)} g(u) \geq 1$ for all $v \in V$. The *fractional total domination number* of a graph G is denoted by γ_{ft} and is given by

$$\gamma_{ft} = \min\{|g| : g \text{ is a total dominating function of } G\},$$

where $|g| = \sum_{v \in V} g(v)$.

This concept of a fractional domination number has a close relation with the magic constant of a graph as stated in the following theorem.

Theorem 1.2. [1, 5] *If a graph G is distance magic, then its distance magic constant $k = \frac{n(n+1)}{2\gamma_{ft}}$.*

Theorem 1.3. [2, 4, 7, 8] *Let G be a distance magic graph on n vertices. If G is distance magic with distance magic labeling f and magic constant k then $\sum_{v \in V} w(v) = nk$.*

Corollary 1.3.1. *If G is an r -regular distance magic graph on n vertices with magic constant k , then $k = \frac{r(n+1)}{2}$.*